How to use ngIf, else, then in Angular By example

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⟵ ngSwitch Directive ngClass Directive ⟶

The `ngIf` is an Angular Structural Directive, which allows us to add/remove DOM Element based on some condition. In this Tutorial let us look at the syntax of ngIf. We will show you how to use ngIf using example code. We will also learn how to use the optional `else` clause & optional `then` clause using the `ng-template`

Applies to: Angular 2 to the latest edition of i.e. Angular 8. Angular 9, Angular 10, Angular 11

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ngIf Syntax

```
<p *ngIf="condition">
    content to render, when the condition is true
</p>
```

The `ngIf` is attached to a DOM element (p element in the above example). `ngIf` is a structural directive, which means that you can add it to any element like `div`, `p`, `h1`, component selector, etc. Like all structural directive, it is prefixed with `*` asterisk

We bind the `*ngIf` to an expression (a condition in the above example). The expression is then evaluated by the `ngIf` directive. The expression must return either true or false.

If the expression evaluates to false then the Angular removes the entire element from the DOM. If true it will insert the element into the DOM.

Hidden attribute Vs ngIf

```
<p [hidden]="condition">
    content to render, when the condition is true
</p>
```

The above achieves the same thing, with one vital difference.

> `ngIf` **does not hide the DOM element. It removes the entire element along with its subtree from the DOM. It also removes the corresponding state freeing up the resources attached to the element.**
>
> `hidden` **attribute does not remove the element from the DOM. But just hides it.**

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The difference between `[hidden]='false'` and `*ngIf='false'` is that the first method simply hides the element. The second method with `ngIf` removes the element completely from the DOM.

By using the **Logical NOT** (!), you can mimic the else condition as shown here.

```
<p *ngIf="!condition">
    content to render, when the condition is false
</p>
```

The better solution is to use the optional `else` block as shown in the next paragraph.

Condition

The condition can be anything. It can be a property of the component class. It can be a method in the component class. But it must evaluate to true/false. The `ngIf` directive tries to coerce the value to Boolean.

ngIf else

The `ngIf` allows us to define optional `else` block using the `ng-template`

```
<div *ngIf="condition; else elseBlock">
    content to render, when the condition is true
</div>

<ng-template #elseBlock>
    content to render, when the condition is false
</ng-template>
```

The expression starts with a condition followed by a semicolon.

Next, we have `else` clause bound to a template named `elseBlock`. The template can be defined anywhere using the `ng-template`. Place it right after ngIf for readability.

When the condition evaluates to false, then the `ng-template` with the name `#elseBlock` is rendered by the `ngIf` Directive.

ngIf then else

You can also define `then` `else` block using the `ng-template`

```
<div *ngIf="condition; then thenBlock else elseBlock">
    This content is not shown
</div>

<ng-template #thenBlock>
    content to render when the condition is true.
</ng-template>

<ng-template #elseBlock>
    content to render when condition is false.
</ng-template>
```

Here, we have `then` clause followed by a template named `thenBlock`.

When the condition is true, the template `thenBlock` is rendered. If false, then the template `elseBlock` is rendered

ngIf Example

Create a new Angular project by running the command `ng new ngIf`

Component Class

Create a boolean variable `showMe` in your `app.component.ts` class as shown below

```
import { Component } from '@angular/core';

@Component({
  selector: 'app-root',
  templateUrl: './app.component.html',
  styleUrls: ['./app.component.css']
})
export class AppComponent {
  title: string = 'ngIf Example' ;
  showMe: boolean;
}
```

Template

Copy the following code to the `app.component.html`.

```
<h1>Simple example of ngIf </h1>


<div class="row">
  Show <input type="checkbox" [(ngModel)]="showMe" />
</div>

<h1>ngIf </h1>

<p *ngIf="showMe">
  ShowMe is checked
</p>
<p *ngIf="!showMe">
  ShowMe is unchecked
</p>

<h1>ngIf Else</h1>

<p *ngIf="showMe; else elseBlock1">
  ShowMe is checked
</p>

<ng-template #elseBlock1>
  <p>ShowMe is unchecked Using elseBlock</p>
</ng-template>

<h1>ngIf then else</h1>

<p *ngIf="showMe; then thenBlock2 else elseBlock2">
  This is not rendered
</p>

<ng-template #thenBlock2>
  <p>ShowMe is checked Using thenblock</p>
</ng-template>

<ng-template #elseBlock2>
  <p>ShowMe is unchecked Using elseBlock</p>
</ng-template>

<h1>using hidden </h1>

<p [hidden]="showMe">
    content to render, when the condition is true  using hidden property binding
</p>

<p [hidden]="!showMe">
    content to render, when the condition is false. using hidden property binding
</p>
```

Now let us examine the code in detail

```
Show <input type="checkbox" [(ngModel)] ="showMe"/>
```

This is a simple checkbox bound to `showMe` variable in the component

```
<div *ngIf="showMe">
    ShowMe is checked
</div>
```

The `ngIf` directive is attached to the `div` element. It is then bound to the expression `“showMe”`. The expression is evaluated and if it is `true`, then the `div` element is added to the DOM else it is removed from the DOM.

```
<p *ngIf="showMe; else elseBlock1">
  ShowMe is checked
</p>

<ng-template #elseBlock1>
  <p>ShowMe is checked Using elseBlock</p>
</ng-template>
```

If else example.

```
<p *ngIf="showMe; then thenBlock2 else elseBlock2">
  This is not rendered
</p>

<ng-template #thenBlock2>
  <p>ShowMe is checked Using thenblock</p>
</ng-template>

<ng-template #elseBlock2>
  <p>ShowMe is unchecked Using elseBlock</p>
</ng-template>
```

If then else example. Note that the content of `p` element, to which `ngIf` is attached is never rendered

```
<h1>using hidden </h1>

<p [hidden]="showMe">
    content to render, when the condition is true  using hidden property binding
</p>

<p [hidden]="!showMe">
    content to render, when the condition is false. using hidden property binding
</p>
```

The property binding on the hidden attribute. You can open the developer console and see that both the Angular renders both the elements. But mark one of them as visible and the other one as hidden.

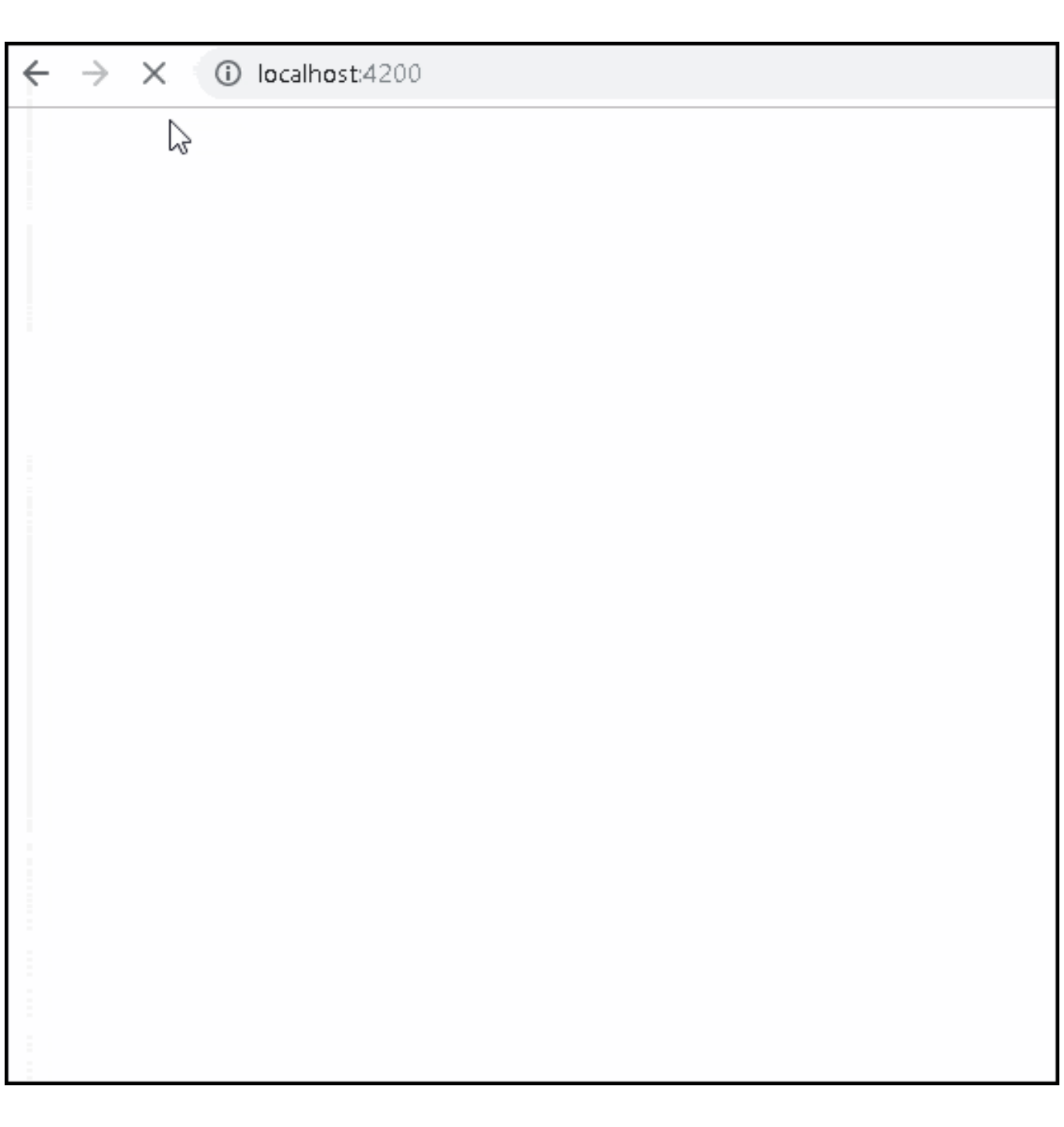
Module

Import FormsModule in app.module.ts as we are using ngModal directive

```
import { BrowserModule } from '@angular/platform-browser';
import { NgModule } from '@angular/core';
import { FormsModule } from '@angular/forms';

import { AppRoutingModule } from './app-routing.module';
import { AppComponent } from './app.component';

@NgModule({
  declarations: [
    AppComponent
  ],
  imports: [
    BrowserModule,
    AppRoutingModule,
    FormsModule
  ],
  providers: [],
  bootstrap: [AppComponent]
})
export class AppModule { }
```



References

ngIf API

Read More

1. Angular Directives
2. ngFor
3. ngSwitch
4. ngIf
5. ngClass
6. ngStyle
7. Angular Track by

Summary

In this tutorial, we learned how to make use of `ngIf` directive to add/remove the DOM element. The `ngIf` can be used along with the optional `then` and `else` clause.

⟵ ngSwitch Directive ngClass Directive ⟶

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